Light

GCSE Computer Science — Data Representation: Media (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

ASCII stands for American Standard Code for Information Interchange.
What type of data does ASCII represent?

- ○ **A.** Images
- ○ **B.** Sound
- ○ **C.** Video
- ○ **D.** Text characters

Q2 (1 mark)

Extended ASCII uses 8 bits per character.
How many different characters can extended ASCII represent?

- ○ **A.** 128
- ○ **B.** 512
- ○ **C.** 1024
- ○ **D.** 256

Q3 (1 mark)

What type of image is made up of a grid of pixels?

- ○ **A.** A bitmap image
- ○ **B.** A text file
- ○ **C.** An audio file
- ○ **D.** A database

Q4 (1 mark)

An image uses 4-bit colour depth.
How many different colours can each pixel be?

- ○ **A.** 4
- ○ **B.** 8
- ○ **C.** 32
- ○ **D.** 16

Q5 (1 mark)

An image is described as having a resolution of 1920 × 1080.
What does 1920 × 1080 represent?

- ○ **A.** The file size in bytes
- ○ **B.** The colour depth
- ○ **C.** The compression ratio
- ○ **D.** The width and height in pixels

Q6 (1 mark)

When sound is converted from analogue to digital, the process involves taking measurements of the sound wave.
What is each measurement called?

- ○ **A.** A pixel
- ○ **B.** A byte
- ○ **C.** A frame
- ○ **D.** A sample

Q7 (1 mark)

CD quality audio uses a sample rate of 44,100 Hz.
What does this mean?

- ○ **A.** 44,100 samples are taken per second
- ○ **B.** The file is 44,100 bytes

- ○ **C.** The bit depth is 44,100
- ○ **D.** The frequency range is 44,100 Hz

Q8 (1 mark)

A digital audio recording uses 16-bit depth.
What does this affect?

- ○ **A.** How often the sound is measured
- ○ **B.** The length of the recording
- ○ **C.** How precisely each sample's amplitude is recorded
- ○ **D.** The file format used

Q9 (1 mark)

Which factor does NOT affect the file size of a bitmap image?

- ○ **A.** The resolution
- ○ **B.** The colour depth
- ○ **C.** The subject of the image
- ○ **D.** The number of pixels

Q10 (1 mark)

ASCII includes codes for control characters.
What is a control character?

- ○ **A.** A letter of the alphabet
- ○ **B.** A number from 0-9
- ○ **C.** A non-printable character like tab or newline
- ○ **D.** A mathematical symbol

Q11 (1 mark)

Increasing the bit depth of a digital audio recording will:

- ○ **A.** Reduce the file size
- ○ **B.** Make the recording longer
- ○ **C.** Improve the audio quality
- ○ **D.** Decrease the sample rate

Q12 (1 mark)

A bitmap image appears blocky and pixelated when enlarged.
This is because:

- ○ **A.** The colour depth is too high
- ○ **B.** The file is corrupted
- ○ **C.** The resolution is too low for the display size
- ○ **D.** The image format is wrong

Q13 (2 marks)

In ASCII, the character 'A' has the denary code 65.
What is the ASCII code for 'G'?

- ○ **A.** 71
- ○ **B.** 70
- ○ **C.** 72
- ○ **D.** 73

Q14 (2 marks)

In ASCII, uppercase 'A' is 65 and lowercase 'a' is 97.
What is the ASCII code for lowercase 'k'?

- ○ **A.** 75
- ○ **B.** 105
- ○ **C.** 107
- ○ **D.** 139

Q15 (2 marks)

An image uses 16-bit colour depth.
How many different colours can be displayed?

- ○ **A.** 256
- ○ **B.** 4,096
- ○ **C.** 65,536

- ○ **D.** 16,777,216

Q16 (2 marks)

A colour image has dimensions of 640 × 400 pixels with 24-bit colour depth.
What is the file size in bits?

- ○ **A.** 6,144,000 bits
- ○ **B.** 7,680,000 bits
- ○ **C.** 10,240,000 bits
- ○ **D.** 15,360,000 bits

Q17 (2 marks)

An audio file is recorded at 48,000 Hz sample rate with 8-bit depth for 2 seconds.
What is the file size?

- ○ **A.** 768,000 bits
- ○ **B.** 384,000 bits
- ○ **C.** 1,536,000 bits
- ○ **D.** 3,072,000 bits

Q18 (2 marks)

A sound is sampled at 4,000 Hz.
What is the sample interval (time between samples)?

- ○ **A.** 0.000025 seconds
- ○ **B.** 0.00025 seconds
- ○ **C.** 0.0025 seconds
- ○ **D.** 0.025 seconds

Q19 (2 marks)

An image has dimensions 2048 × 1536 pixels and uses 8-bit colour depth.
What is the file size in bytes?

- ○ **A.** 393,216 bytes
- ○ **B.** 3,145,728 bytes
- ○ **C.** 6,291,456 bytes
- ○ **D.** 25,165,824 bytes

Q20 (2 marks)

A digital audio recording of a symphony orchestra sounds 'flat' and lacks the subtle variations in the original performance.
What is the most likely cause?

- ○ **A.** The sample rate is too high
- ○ **B.** The bit depth is too low
- ○ **C.** The file format is incorrect
- ○ **D.** The recording is too long

Q21 (2 marks)

Which audio settings will produce the smallest file size for a 10-second recording?

- ○ **A.** 48,000 Hz sample rate, 16-bit depth
- ○ **B.** 22,050 Hz sample rate, 16-bit depth
- ○ **C.** 44,100 Hz sample rate, 8-bit depth
- ○ **D.** 11,025 Hz sample rate, 8-bit depth

Q22 (2 marks)

The ASCII code for 'P' is 80 in denary.
What is this in binary?

- ○ **A.** 1001110
- ○ **B.** 1010000
- ○ **C.** 1010010
- ○ **D.** 1010100

Q23 (2 marks)

An artist's digital painting shows visible 'stepping' in areas where colours should smoothly blend together.
What is this limitation called and what causes it?

- ○ **A.** Pixelation, caused by low resolution
- ○ **B.** Aliasing, caused by diagonal lines

- ○ **C.** Colour banding, caused by limited colour depth
- ○ **D.** Compression artefacts, caused by file format

Q24 (3 marks)

A smartphone takes photos at 3264 × 2448 pixels with 24-bit colour. How many photos can fit on a 2 GiB memory card?
(1 GiB = 1024^3 bytes)

- ○ **A.** Approximately 45 photos
- ○ **B.** Approximately 89 photos
- ○ **C.** Approximately 178 photos
- ○ **D.** Approximately 356 photos

Q25 (3 marks)

A stereo audio file is recorded at 48,000 Hz sample rate with 24-bit depth for 4 minutes.
What is the approximate file size in MiB?
(Stereo = 2 channels, 1 MiB = 1024^2 bytes)

- ○ **A.** Approximately 33 MiB
- ○ **B.** Approximately 55 MiB
- ○ **C.** Approximately 66 MiB
- ○ **D.** Approximately 110 MiB

Q26 (3 marks)

Image X is 1024 × 768 with 8-bit colour. Image Y is 512 × 384 with 32-bit colour.
Which statement correctly compares their file sizes?

- ○ **A.** Image X is twice as large as Image Y
- ○ **B.** Image Y is twice as large as Image X
- ○ **C.** They are exactly the same size
- ○ **D.** Image X is four times as large as Image Y

Q27 (3 marks)

The word 'Soda' is stored using ASCII.
If 'S' = 83, what is the total of all ASCII codes for 'Soda'?

- ○ **A.** 391
- ○ **B.** 393
- ○ **C.** 395
- ○ **D.** 397

Q28 (3 marks)

Recording P uses 48,000 Hz sample rate and 8-bit depth. Recording Q uses 24,000 Hz sample rate and 16-bit depth.
Both recordings are 5 minutes long. Which statement is correct?

- ○ **A.** Recording P is larger than Recording Q
- ○ **B.** Recording Q is larger than Recording Q
- ○ **C.** Both recordings are exactly the same size

- ○ **D.** Recording P captures higher frequencies than Recording Q

Q29 (3 marks)

A digital pressure sensor uses 10-bit values, where 0 represents 0 kPa and 1023 represents 500 kPa.
What is the precision (smallest detectable change) of this system?

- ○ **A.** Approximately 0.25 kPa
- ○ **B.** Approximately 0.49 kPa
- ○ **C.** Approximately 0.98 kPa
- ○ **D.** Approximately 1.96 kPa

Q30 (3 marks)

A game developer needs to reduce sprite images from 1 MiB to under 128 KiB each without changing dimensions.
The sprites are currently 512 × 512 pixels with 32-bit colour.
What is the maximum colour depth that meets this requirement?

- ○ **A.** 8-bit colour
- ○ **B.** 4-bit colour
- ○ **C.** 2-bit colour
- ○ **D.** 1-bit colour

Total: **55** marks • Time: **30 min**

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